

Chapter 7: Planning a GIS Project

1. Introduction to GIS Project Planning

A GIS project is a structured, goal-oriented endeavor designed to solve a specific spatial problem or support a particular management decision. Because GIS integrates complex hardware, multi-layer databases, specialized software, and skilled personnel, a project's success depends far more on systematic, strategic planning than on the software tools used.

As emphasized by **George B. Korte**, a project that lacks clear planning will suffer from data redundancy, budget overruns, and "scope creep" (uncontrolled project expansion). A structured design ensures that the data collected matches the exact spatial accuracy required for the final analysis.

2. Structural Steps in a GIS Project Lifecycle

While different institutional workflows vary slightly, the classic methodology laid out by **Kang-tsung Chang** divides a GIS project into five distinct, sequential phases:

1. Problem Identification → 2. Data Acquisition & Structuring → 3. Spatial Analysis Engine → 4. Evaluation & Verification → 5. Result Presentation

Step 1: Problem Identification and Objective Definition

- **Scope Formulation:** Clearly define the real-world spatial problem to be solved (e.g., "Selecting the optimal site for a new regional landfill in the district").
- **Criteria Determination:** Establish the strict geographic and environmental rules that must guide the project (e.g., *Must be located on flat slopes, must not overlap with wetlands, and must be at least 2 kilometers away from residential areas*).
- **Target Audience Specification:** Identify who will use the final results to ensure the data is presented at an appropriate technical level.

Step 2: Data Acquisition, Quality Control, and Database Design

- **Inventorying Data Requirements:** Identify what specific spatial layers are needed (e.g., Digital Elevation Models, soil maps, road networks, parcel registries).
- **Format and Coordinate Harmonization:** Acquire data from primary or secondary sources and ensure all layers are explicitly projected to the *same* local Coordinate Reference System (CRS) and map datum to prevent overlay misalignment.
- **Data Editing and Cleaning:** Run topological error checks to clip out dangles, close incomplete polygons, and snap overlapping vertices.

Step 3: Performing the Spatial Analysis

- **Workflow Construction:** Execute the structured spatial commands (e.g., buffer generations, raster map algebra queries, vector intersections) to isolate areas that match the project criteria.
- **Intermediary Layer Management:** Organize and track temporary data layers generated during multi-step analytical pipelines.

Step 4: Evaluation and Validation

- **Sensitivity Analysis:** Check how minor changes to the input criteria (e.g., widening a protective river buffer from 100 meters to 150 meters) alter the final result.
- **Ground-Truthing:** Validate the computer-generated outputs against real-world field conditions or satellite imagery to verify structural accuracy.

Step 5: Presenting and Implementing Results

- **Deliverable Compilation:** Transform abstract spatial database rows into human-readable map layouts, interactive web dashboards, or text summaries for decision-makers.

3. Performing the Analysis (Methodology Formulation)

The core analytical phase must convert the real-world project criteria into an explicit, step-by-step **spatial modeling flowchart** before opening the GIS software. This prevents computational trial-and-error.

Organizing a Selection Matrix (Example: Health Clinic Site Selection)

To run an analysis systematically, criteria are translated into measurable GIS commands:

Real-World Objective/Criteria	Spatial Analysis Method Used	Target Layer Output Type
1. Must be close to existing roads	Vector Buffer (e.g., 500m) around road lines)	Proximity Polygon
2. Must avoid steep terrain hazards	Slope Derivation from a DEM (extract slopes (10%))	Suitable Slope Grid
3. Must be built on vacant public land	Attribute Query (SELECT WHERE Status = 'Vacant')	Available Land Vector
4. Find the final overlapping zones	Overlay Intersection (Combine steps 1, 2, and 3)	Final Candidate Parcels

4. Presenting the Results

As **Stan Aronoff** notes from a management perspective, the ultimate power of a GIS is its ability to turn raw geospatial data into an actionable format for stakeholders who may not understand GIS software. The results must be presented using clear, communicative formats:

A. Cartographic Design (The Professional Map Layout)

Maps remain the primary communication tool in geography. To be effective, every final layout must contain these fundamental cartographic elements:

- **Thematic Symbology:** Using clear, intuitive colors and symbols (e.g., using red gradients for high-hazard areas, blue for hydrography, and green for conservation zones).

- **Directional Orientation:** A clear North Arrow or coordinate grid system to anchor the map geographically.
- **Scale Bars:** Both graphic scale bars and numeric text ratios (\$1:50,000\$) to allow users to make accurate physical distance estimates.
- **Comprehensive Legend:** A clean key that decodes every single layer symbol, color, or hatch pattern used on the map.
- **Metadata Metadata Elements:** Incorporating titles, publication dates, author credits, and official citations of the underlying raw data sources to ensure transparency and accountability.

B. Tabular Summaries and Analytical Reports

- Visual maps should be accompanied by numerical reports that summarize the findings (e.g., a spreadsheet layout detailing the total square kilometers of land suitable for development, broken down by district ownership).

C. Digital Formats and Web Deployment

- **Geospatial Files:** Exporting final project data into standardized, cross-platform industry formats (such as Esri Shapefiles, GeoTIFFs, or GeoJSON strings).
- **Web Mapping Dashboards:** Publishing interactive web map services (e.g., via ArcGIS Online, QGIS Cloud, or custom Leaflet web layouts) that allow clients or the general public to pan, zoom, and query attribute data directly from a browser.